

Research Paper

Recommended citation: Olsson, T. (2025). Ethics For Design Students: A Framework of Ethical Competence to Scaffold Course Planning . In Kangaslampi, R., Langie, G., Järvinen, H.-M., & Nagy, B. (Eds.), SEFI 53rd Annual Conference. European Society for Engineering Education (SEFI), Tampere, Finland. DOI: 10.5281/zenodo.17631762.

This Conference Paper is brought to you for open access by the 53rd Annual Conference of the European Society for Engineering Education (SEFI) at Tampere University in Tampere, Finland. This work is licensed under a Creative Commons Attribution-NonCommercial-Share Alike 4.0 International License.

ETHICS FOR DESIGN STUDENTS: A FRAMEWORK OF ETHICAL COMPETENCE TO SCAFFOLD COURSE PLANNING

TD Olsson

Tampere University, Tampere, Finland, https://orcid.org/0000-0002-1106-2544

***Conference Key Areas**: Sustainability and society in engineering, Engineering ethics education*
***Keywords**: Digital ethics, Design, Professional responsibility, Sustainable development, Ethical competence, Maturity model*

ABSTRACT

The necessity to educate applied ethics to engineering students is well established in literature but the practice in higher education is trailing behind. A key question is how to introduce ethical contents relevant to specific fields and in pedagogically effective ways. This article focuses on students of human-centric design in the context of software development, a diverse group of future experts who ought to address complex human concerns in software systems through responsibilities like requirements engineering, service design and user experience design. The article presents a framework of developing designers' ethical competence at four tiers (awareness, responsibility, practice, methods) and offers examples of relevant contents and pedagogical choices to support learning. The framework is based on the author's experiences of developing a course for a multi-disciplinary and multicultural audience of Master students in human-computer interaction and IT for sustainable development. It is intended to help scaffold course planning primarily for design students, but may also inform ethics education in computer science, software production and engineering in general.

1 INTRODUCTION AND RELATED WORK

Over the past decade, the broad community in computer science (CS) and software engineering (SE) has been undergoing a process that could be seen as an *ethical awakening*. Academics as well as practitioners are increasingly aware of the diverse effects of digitalization—both the beneficial and harmful, intended and unintended—on individuals (e.g., Lunn et al., 2025), as well as cultures, societies and the environment (e.g., Crawford, 2021). This has intensified the discussion related to the social and political power of computing technology (Beer, 2017) and the CS community's responsibility to society and its values and moral grounds (e.g., Chance et al., 2025). Especially the abundant discussion about artificial intelligence, biasfree algorithms and lack of transparency has raised the intensity of so-called *techno-ethical issues* (Shilton, 2018). Taking a historical perspective, it could be said that the techno-ethical dimension is entering the stage of software engineering as firmly as agile development, lean thinking or user experience did some 20 years ago.

This trend in SE practice consolidates the much-discussed need for bringing techno-ethical questions more to the forefront in CS and SE education, as well as design and engineering in general. It is no longer a question of *whether or not* we should teach ethics in engineering (e.g., Chance et al., 2025), but the question is *how* we can effectively integrate relevant elements into higher education curricula (Polmear et al., 2025). This growing demand has motivated research on suitable pedagogical approaches and techniques in teaching engineering ethics (Barakat et al., 2023). For example, Duarte and others (2019) call for problem-based learning and multicultural and multidisciplinary teamwork to reach good learning effects, supported by Lundqvist's (2016) argument to integrate ethics through practical cases. Børsen et al. (2021) suggests involving students in the teaching by utilizing methods of active pedagogy and by linking the content close to real life. These approaches appear pedagogically promising, yet the question of how to scaffold the development of individuals' ethical competences remains to be addressed by theory and empirical trials. For example, Polmear et al. (2025) underline the gap of studying the non-cognitive aspects of ethics education (e.g., emotional and motivational engagement), which could help understand why people might act unethically without meaning to.

This article addresses these pedagogical challenges by considering "digital ethics" (Ashok et al., 2022) education in the context of *design* in SE—typically termed Human-Centric Design (HCD). The roles of HCD and engineering typically overlap significantly in SE practice, but certain fundamental differences can be identified. While engineering roles tend to deal with questions of *how*, that is, seeking algorithmic solutions to, for example, search engine bias or automated decision-making (O'Sullivan et al., 2023), design focuses more on the question of *why*. People identifying as designers play key roles in defining the very *raison d'être*, the *telos* of software artefacts; they are expected to consider questions like what purposes software applications are developed for and what their intended effects are. Like in any field of design, also HCD is concerned with *change and preferred futures*: "Design is the ability to imagine that-which-does-not-yet-exist, to make it appear in concrete form as a new purposeful addition to the real world" (Nelson & Stolterman, 2012). Those who define the visions, goals and key features of software also set conditions for users' experiences and behaviour for years to come. Indeed, techno-ethical dilemmas often relate to the purposes of and ways of using IT applications, rather than to how they have been technically engineered. Design can

be seen as a mediator between the complex human requirements and the mathematical formalism of SE, responsible for the translation of human concerns into designable features. Mindful of this inherent power, designers should also take the responsibility of reconsidering the practices and values that shape the ethical quality of software through design activities. In this light, it is easy to agree with Cennydd Bowles's dictum *"Design is applied ethics"* (2018). Against this backdrop, we argue for a need of diverse theoretical frameworks to cater for the particularities of specific professional practices—HCD in this case.

This article further argues that educational development needs to support individuals building up a *general sensitivity* and consideration of the ethical nature of IT and higher-level understanding of the role of engineering and design in society (e.g., Zhu et al., 2025). For example, Sætra & Danaher (2022) argue that there is no need to create a separate "ethics of X" for each subtype of technology (e.g., computer ethics, AI ethics, robot ethics). While specific technologies and applications can have specific impacts, the ethical issues may be sufficiently covered and understood through already established *higher-level domains of ethics*. An increasingly prevalent example of such higher-level domain is the notion of *sustainability*. Sustainability underlines the long-term ideal of balanced co-existence of the human and natural worlds, mindful of both environmental, social and economic perspectives (Purvis et al., 2019). A growing body of research also in the SEFI community highlights the essence of sustainability training in engineering higher education (e.g., Tepsa et al., 2023; Klonowska et al., 2023). Sustainable IT should, for example, nurture responsible consumption behaviour in the digital realm, avoid actions that endanger the environment through disproportional use of energy for computing and data transfer and avoid undermining the autonomy and privacy of individual users. Focusing on sustainability as another broad umbrella concept highlights the need for IT to broaden the notion of quality—expanding from the user-centric and individual perspectives typically considered in HCD to systemic considerations like cultural and environmental effects. The lens of sustainability can thereby help educate "engineers to be socially responsible and ethically aware professionals who can contribute to achieving sustainable development goals" (Tepsa et al., 2023).

Against this backdrop, this article outlines a theoretical framework for developing ethical competence in digital ethics education in the context of human-centric design for sustainable development (see Fig. 1, explained in Section 3). The framework is aimed to support course planning by introducing four tiers of digital ethics. Due to the context in which it was created, the primary target group is students pursuing design-related roles, such as service designers, user experience designers, information architects and product owners. In what follows, I describe the starting points and main steps of developing the framework.

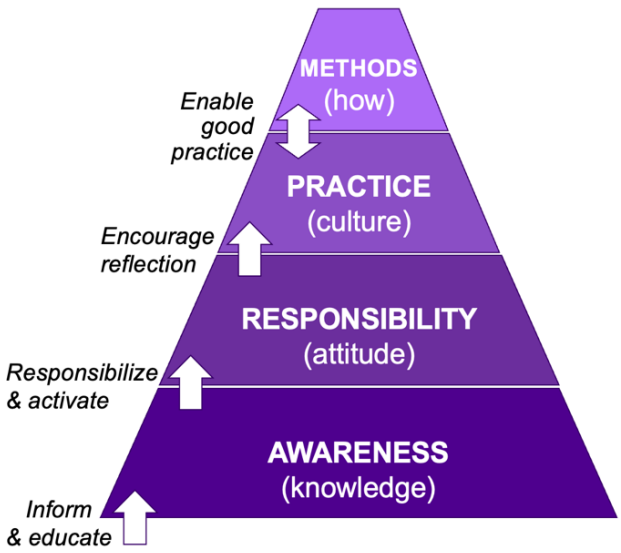


Fig. 1. Framework of ethical competence: four tiers to consider in design ethics education.

2 METHODOLOGY OF THE FRAMEWORK DEVELOPMENT

This theoretical work draws from several areas of literature, as well as from personal pedagogical experience and reflections on the software industry. The process of developing the framework is characterized by personal reflection, iterative development and a bit of serendipity, rather than by a systematic methodology or step-by-step process. To support interpretation, the following sheds light on the contextual and disciplinary underpinnings that have influenced its current form.

First, it is key to understand that my disciplinary background is in education of Human-Computer Interaction (HCI) and Human-Centric Design in particular. In my operating environment at Tampere University, Finland, these are closely tied to education of SE on the one hand and design thinking and user research education on the other. The activity focus of the course is on design of software products and services, covering practices like concept design, service design, requirements engineering, user experience design and information architecture. Second, the framework has developed iteratively over five years of giving a new course (called Sustainable Design) and continuously refining it for the needs of a cross-disciplinary Master's degree programme, a more classical HCI programme and an accessibility programme. Rather than focusing on, for example, planning ethics modules for first-year students (e.g., Amashi et al., 2021), the course was aimed at quite experienced MSc students, most of them having practical experience in work life. The first version was developed already at the course planning stage and was mainly driven by pedagogical intuition rather than specific theories or empirical data. Over time, the framework played key role in refining the course learning goals and pedagogical choices. The refinement has been informed by qualitative course feedback as well as my personal reflections as an educator. At this stage of research, there is no hard evidence about the educational impact of the framework. However, certain weak signals exist, implying that follow-up studies on its effectiveness could be worthwhile: (1) generally highly positive course feedback through qualitative remarks in students' self-assessments and the summative course feedback (overall grade above 4.5, on a 1–5 scale); (2) growing attractiveness of the course (student numbers per implementation have grown from approx. 30 to 40, 60 and most recently over 90).

3 FRAMEWORK FOR SCAFFOLDING COURSE PLANNING

The framework is intended to help consider, plan and assess students' so-called ethical competence (Zhu et al., 2025). Figure 1 visualizes the framework as a pyramid with four tiers, spanning from *epistemic* aspects (awareness) to *behavioural attitudes* (responsibility as a mindset), *resulted action* at a cultural level (practice) and finally the *tools and methods* that support ethical thought and its cultivation in day-to-day teamwork. The overall structure reflects certain elements from the Bloom's taxonomy and its levels of learning (1956), particularly by highlighting knowledge as the foundation. However, this framework does not focus on individual learners' depth in specific topics but on the extent to which ethical sensitivity and consideration is manifest in their thoughts, intentions, actions and practice. The tiers on the pyramid are meant as qualitative perspectives to how ethical consideration manifests. At the level of organisations or developer teams, the framework could also be used as a high-level *maturity model* to help assess collective's capabilities in this regard. The following sections unpack each tier with examples of relevant contents and outlines what competences are beneficial for reaching the next tiers.

3.1 Awareness Building

The following outlines aspects and issues that I consider relevant to build awareness of the ethical nature and power of IT and possible related harms (Lunn et al., 2025), helping to sensitize students to potential ethical issues (Barakat et al., 2023) and practice systems thinking. First, a learner needs to realize and analyse various problematic examples where IT artefacts have resulted in significant and diverse forms of harm. Due to the extent of related harms, it is key to select examples that are professionally relevant, timely and present relatable forms of harm and power—for example, by recognizing the effects in their own user experiences of consumer applications. My opening lectures have started this problematization with a classic example: the mechanism of crowdsourced popularity ranking on most web services, such as restaurant reviews and product reviews on online stores. Students can recognize both the benefits of such a simple UI-level decision (e.g., customer-centricity and high transparency) and the harms from anonymous users being given the power to publicly praise or shame service providers, and how this can support dehumanizing forms of transactional interactions and the creation of a network of peer surveillance, voluntarily and for free (criticized by, e.g., Harari, 2024). Other examples with problematic aspects range from mundane UI features like Autoplay on social media (maximizing user engagement) to more timely and computationally sophisticated applications like Generative AI or Deepfake video manipulation.

Second, students need to become aware of the more general mechanisms of harm in IT and design. To this end, Nelson and Stolterman's (2012) conceptualization of *evil* in design is very useful: design can be evil when it results in something undesirable that becomes part of the world. They distinguish between natural, wilful and accidental evil, covering both intended and unintended as well as expected and unexpected forms of harm resulting from design choices. Often harm emerges in long term and unexpectedly by misfortune, accident or by using the designer's power without understanding the possible environmental or societal repercussions. Third, considering sustainability as a lens to the systemic quality of IT, it is important to understand the traditional pillars of sustainability (environmental, socio-cultural, economic) and the different temporal horizons they focus on—from the millennial effects on environment to decennial socio-cultural and economic influences. Operating with a multi-layered concept like sustainability encourages systems thinking and broadening the perspective from human-centric quality criteria to system-level criteria. Furthermore, applied ethics in other emerging technological domains like healthcare and biotechnology (Sandler, 2014) can showcase the systemic complexities and moral issues with technology in other fields.

Finally, societal level of knowledge pertains to social scientific truisms about technological change and disruptions. After the aforementioned concrete examples, it is helpful to understand various general and long-debated standpoints concerning the interrelations between technology and culture. For example, Melvin Kranzberg's laws of technology (1986) underline that technologies are non-neutral: while the morality of a technological application is partly defined by its uses, also the capability to afford certain malicious uses demands attention. Similarly, in his 1998 talk, Neil Postman argued, for example, that the benefits and costs of technology are never distributed evenly among the population, and that technological change is neither additive nor incremental but ecological and systemic. James Moor's (1985) Law "as technological revolutions increase their social impact, ethical problems increase"

reminds us of the fundamentally ethical nature of any technology. These views echo Braun and others' (2025) argument about the importance of expanding the basis of ethical discussion to diverse social scientific views (e.g., science and technology studies, critical theory) as "engineers are also tacit sociologists".

3.2 Nurturing Responsibility

The abovementioned forms of awareness allow self-reflection and sensitivity to the power of design over other people, hence cultivating budding sense of *responsibility*. Professional responsibility is a key question in technology ethics in terms of dealing with accountability for negative impacts of engineering on society (Luppicini, 2009). To operationalize such a broad concept, responsibility can be considered at three levels (Herwix et al., 2022). Responsibility at the *individual level* is concerned with the professional responsibility of designers, engineers and other professionals in the development of IT-based solutions (Ibid.). This relates to what Zhu et al. (2025) call "character building" (personal traits, habit-like tendencies). To this end, codes of ethics, integrity guidelines and professional manifestos are generally used to establish widely shared norms and ideals that professionals in a specific field should respect. At the *organizational level*, responsibility can be taken in more holistic ways, covering pluralistic viewpoints to the systemic quality of software. Relevant issues at this level include the Problem of the Many Hands, referring to problems related to individual accountability and the ascription of responsibility in collective settings (Franssen et al., 2018). For example, organisational responsibility can suffer from unclear roles and agency, power vacuums, unclear leadership structures or a lack of speak-up culture. Finally, responsibility *at the cultural level* is influenced by the narratives, paradigms and general attitudes in the professional community. These are unpacked in the next subsection about practice and professional culture.

To elaborate on different notions of responsibility, I find it important to offer an overview of the various normative ethics theories and to recognize more long-standing discourses in philosophy (e.g., Shilton, 2018). These include, for example, the tensions between *consequentialist ethics* (goodness of action being defined by the consequences) vs. *deontological ethics* (or ethics of duty, underlining the need for establishing ground rules or universal obligations that must be followed in all cases). On the other hand, Aristotelian *virtue ethics* and the more modern notion of *ethics of care* can help cultivate professional responsibility on personal level. Ethical competence necessitates understanding and clarifying both *values*, as the ends of design, and *virtues*, as the means to reach them through wise thinking and action. Therefore, course assignments and group discussions should be arranged such that a learner is exposed to taking different values, virtues and theories of normative ethics as the vantage points to design thinking.

As responsibility is also connected to *identity*, it is important to shape the learners' professional sense of self and help clarify their personal values. This can include repeated discussions on the roles and responsibilities of designers (e.g., related to the problematic examples covered under Awareness Building). This can also include empowering discussions that encourage positive role models and build skills in continuous self-reflection. For example, underlining the potential for good in a new generation of designers can motivate to reconsider professional conventions and build courage to defend the relevance of moral thought as part of design and SE.

3.3 Influencing Practice and Professional Culture

A sense of responsibility and clarity of values are key building blocks for ethical professional practice. However, when moving from individuals to the collective level, it is important to mind certain pitfalls that can prevent the realization of individuals' good intentions. For example, the *talk–do gap* refers to challenges in putting the discussed issues into practice due to challenges in agreeing on how to work in practice. The mundane and repetitive nature of professional practice can also lead to *akrasia*, i.e., lack of self-control and acting against one's better judgment (as hinted also by Polmear et al., 2025).

The CS field involves many cultural influences, conventions and biases that shape the general ways of thinking and values of individual practitioners. These introduce *pitfalls* that can undermine ethical consideration and should therefore be questioned and addressed (Olsson, 2025). For example, *techno-solutionism* is prevalent in CS and refers to a way of thinking that all problems can be addressed by technology—a "confidence in the human capacity to find and implement numerous strategies for overcoming obstacles and capitalizing on opportunities" (Sætra & Selinger, 2023). On the one hand, it reflects a hopeful belief that all difficulties have benign solutions, and that computation offers a potent solution with minimal harm; on the other hand, it is a learned attitude towards designers' omnipotent capabilities to address any problems, which can lead to over-confidence and bold attempts to digitalise anything, anywhere (Olsson, 2025). Another pitfall is the *grand narratives* (Saleebey, 1994) that echo common beliefs, ideals, expectations and an overall ethos in the professional culture (Olsson, 2025). The collective mentality in SE and related business could be simplified into catchphrases like "*Move fast and break things; unless you are breaking stuff, you are not moving fast enough"* or "*Software is eating the world"*. These reflect the ideals of agility, radical innovations, trial-and-error mindset, a growth-driven mindset and an idolisation of the culture of technology-driven innovation (Ibid.). This is often backed by *technological determinism* that espouses the idea of technical progress following a unilinear course and an exogenous force shaping society, contrasting with view that cultural ideas and values shape both technologies and the people creatively appropriating them (Feenberg, 2010). Such innovation-led, disruption-centric narratives can include pathological elements when the morals, values and long-term aims of the practitioners remain unclear.

To avoid the pitfalls, it is key that the established practices and processes encourage critical thinking, reflexive production processes and continuous consideration of the ethical dimensions. In practice, these can include practices like conducting design critique in the form of pre/post-mortems, as well as consideration of the onboarding process and personnel training programs, which often shape the value basis and day-to-day practices of design teams. Overall, this tier calls for an organisational context, which means that a single course has limited ways to expose learners to such questions, other than through group assignments and realistic design projects.

3.4 Methods and Tools to Concretize and Contextualize

Finally, to enable the previous tiers in practice, one needs a collection of methods and tools to concretize the day-to-day work in design projects. These can include artefacts that serve as reminders about ethical perspectives (e.g., stakeholders, forms of impact, intended vs. unintended benefits and harms) or that support

reflection and risk prevention. A sense of responsibility and ethical sensitivity can be gradually built up by employing a broad range of tools throughout a course in both lectures and practical assignments.

First, card deck methods like the Envisioning cards from Value-Sensitive Design (Friedman & Hendry, 2012) and Tarot cards of Technology[1] support perspective taking and empathizing with potential stakeholders and help identify possible benefits and harms for them. Countless other inspirational card decks can be found in design and innovation toolkits, offering playful approaches to nurture systems thinking and lateral thinking. Second, various design canvases help brainstorm and arrange the analysis both visually and collaboratively, typically supported by step-by-step guidance to lower the threshold of starting the work. Examples include Moral Value Map and Normative Design Scheme by ethicsfordesigners.com and the Digital Ethics Canvas by Hardebolle et al. (2023). In addition to such structured procedures, various collaborative methods and practices can support prefiguration and foresight, underlining the importance of collaborative action. For example, speculative risk analysis can be performed with pre-mortems as a form of internal review, where a project team imagines a scenario where the project has failed and then works backwards to determine what could have led to the failure. Finally, professional ethics guidelines serve as general guidance to nurture a sense of professional responsibility. An example of a detailed set of guidelines is the Web Sustainability Guidelines[2], while professional associations like ACM, UXPA and IGDA have outlined more general professional virtues and norms to aspire.

All in all, such methods help to focus on specific contexts; otherwise, ethical consideration can remain too abstract or vague for practically oriented designers and developers. Many of the tools are rather simple as artefacts but can be highly effective in terms of supporting perspective taking, teamwork, slowing down and empathy building. That said, it is key to regard them as just the surface of ethical competence; focusing on them alone introduces a risk of blindly trusting the tools and externalizing judgement and thinking to the procedures they define.

4 CONCLUDING REMARKS

The presented framework aims to support individuals and teams building up a *general sensitivity and consideration* of the ethical nature of software and a higher-level understanding of the role of engineering and design in society. It draws from applied ethics, systems thinking and sustainability science and translates social scientific insights into the practice of software development and human-centric design. For now, it is to be read as a preliminary framework that calls for refinement and validation through empirical studies. While the primary focus is on design education, the framework could inform educational development also for other roles in CS and SE, including software architects, developers and data scientists. I hope the framework can inform and inspire development of digital ethics education that truly supports designers and developers in their ethical sensitivity and consideration and regard the ethical perspective as highly useful for the long-term success and sustainability of all software systems.

[1] https://tarotcardsoftech.artefactgroup.com/
[2] https://sustainablewebdesign.org/guidelines/

REFERENCES

Amashi, R., Joshi, G. & Kandakatla, R. (2021). The influence of sustainable development module on the values and beliefs of first-year students in undergraduate engineering education. *Journal of Engineering Education Transformations 34: 98-103.* https://dx.doi.org/10.16920/jeet/2021/v34i0/157112

Ashok, M., Madan, R., Joha, A., & Sivarajah, U. (2022). Ethical framework for Artificial Intelligence and Digital technologies. *International Journal of Information Management, 62,* https://doi.org/10.1016/j.ijinfomgt.2021.102433.

Barakat, N., Børsen, T., Hardebolle, C., Kovacs, H., Martin, D., & Väätäjä , H. (2023). Techniques In Teaching Engineering Ethics and Professional Responsibility. European Society for Engineering Education (SEFI). https://doi.org/10.21427/9Z3V-QG13

Beer, G. (2017). The Social Power of Algorithms. *Information, Communication and Society,* pp. 1-13.

Bloom, B.S., Engelhart, M.D., Furst, E.J., Hill, W.H., & Krathwohl, D.R. (1956). Taxonomy of educational objectives: The classification of educational goals. *Vol. Handbook I: Cognitive domain.* New York: David McKay Company.

Bowles, C. (2018). *Future Ethics.* NowNext Press, 230p.

Børsen, T., Serreau, Y., Reifschneider, K., Baier, A., Pinkelman, R., Smetanina, T., & Zandvoort, H. (2021). Initiatives, experiences and best practices for teaching social and ecological responsibility in ethics education for science and engineering students. *European Journal of Engineering Education 46(2)*, pp. 186-209. https://doi.org/10.1080/03043797.2019.1701632

Braun, R., Kleba, J., Randell, R. (2025). Sociological, Postcolonial, and Critical Theory Foundations of Engineering Ethics Education. *In: Chance, S. et al. (Eds) The Routledge Interactional Handbook of Engineering Ethics Education.* London: Routledge. p. 165–179. https://doi.org/10.4324/9781003464259

Chance, S., Børsen, T., Martin, D.A., Tormey, R., Lennerfors, T.T., & Bombaerts, G. (Eds.). (2025). *The Routledge International Handbook of Engineering Ethics Education (1st ed.).* London: Routledge. https://doi.org/10.4324/9781003464259

Crawford, K. (2021). *Atlas of AI: power, politics, and the planetary costs of artificial intelligence*. New Haven: Yale University Press.

Duarte, A.J., Malheiro, B., Arnó, E., Perat, I., Silva, M.F., Fuentes-Durá, P., Guedes, P. & Ferreira, P. (2019). Engineering education for sustainable development: the European project semester approach. *IEEE Transactions on Education 63, no. 2,* pp. 108-117. https://doi.org/10.1109/TE.2019.2926944

Feenberg, A. (2010). *Between reason and experience: Essays in technology and modernity.* MIT Press. https://doi.org/10.7551/mitpress/8221.001.0001

Franssen, M., Lokhorst, G.-J., & van de Poel, I. (2018). Philosophy of technology. *In E. N. Zalta (Ed.), The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University.

Friedman, B. & Hendry, D. (2012) The envisioning cards: a toolkit for catalyzing humanistic and technical imaginations. *Proc. of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '12). https://doi.org/10.1145/2207676.2208562

Harari, Y.N. (2024). *Nexus: A Brief History of Information Networks from the Stone Age to AI.* Random House. 528p.

Hardebolle, C., Macko, V., Ramachandran, V., Holzer, A., & Jermann, P. (2023). Digital Ethics Canvas: A Guide for Ethical Risk Assessment and Mitigation in the Digital Domain. European Society for Engineering Education (SEFI). https://doi.org/10.21427/9WA5-ZY95

Herwix, A., Haj-Bolouri, A., M. Rossi, M., Tremblay, S., Purao, & S. Gregor (2022). Ethics in Information Systems and Design Science Research: Five Perspectives. *Communications of the Association for Information Systems, 50.* https://doi.org/10.17705/1CAIS.05028

Klonowska, K., Teljega, M., & Einarson, D. (2023). A Three-Year Academic Track Towards Literacy In Sustainable Development - A Computer Science Study Program Case. European Society for Engineering Education (SEFI). DOI: 10.21427/CFDR-MX42

Kranzberg, M. (1986). Technology and History: "Kranzberg's Laws". *Technology and Culture*, 27(3): 544–560. https://doi.org/10.2307/3105385

Lundqvist, U. (2016). Experiences in integrating ethics for engineers in MSc programmes. *Proceedings of the 12th International CDIO Conference*. 397.

Lunn, S.J., Hazewindus, I., Prasad, P., Ramachandran, V. (2025). Ethical Issues in Software Engineering. *In: Chance, S. et al. (Eds) The Routledge Interactional Handbook of Engineering Ethics Education*. London: Routledge. p. 321–338. https://doi.org/10.4324/9781003464259

Luppicini, R. (2009). The Emerging Field of Technoethics. In: Luppicini, R., & Adell, R. (Eds.), *Handbook of Research on Technoethics*, pp. 1-19. IGI Global. http://doi:10.4018/978-1-60566-022-6.ch001

Moor, J.H. (1985). “What is computer ethics?” *Metaphilosophy. 16(4),* pp. 266–275. http://doi.org/10.1111/j.1467-9973.1985.tb00173.x.

Nelson, H.G., & Stolterman, E. (2012). *The design way: Intentional change in an unpredictable world*. MIT Press. https://doi.org/10.7551/mitpress/9188.001.0001

Olsson, T. (2025, *to appear*) Understanding Moral Lapses in Design: Cultural Biases, Beliefs and Narratives as Factors of Accidental Evil. In: Dovbysh, O., Laaksonen, S., & Pantti, M. (Eds.) *Platforms and the Planet: Big Tech, Digital Platforms and Environmental Responsibility*. Emerald Publishing, 2025.

O'Sullivan, D., Gibson, J. P., Curley, A., Becevel, A., Murphy, E., & Gordon, D. (2023). Ethics4eu: Designing New Curricula for Computer Science Ethics Education: Case Studies for AI Ethics. European Society for Engineering Education (SEFI). https://doi.org/10.21427/5YYB-6H60

Polmear, M., Børsen, T., Love, H.A., Hedayati, A. (2025). Literature Review of Teaching Methods: Trends and Ways Forward to Support Engineering Ethics Instruction. *In: Chance, S. et al. (Eds) The Routledge Interactional Handbook of*

Engineering Ethics Education. London: Routledge. p. 345–362. https://doi.org/10.4324/9781003464259

Purvis, B., Mao, Y. & Robinson, D. (2019). Three pillars of sustainability: in search of conceptual origins. *Sustain Sci 14*, pp. 681–695. https://doi.org/10.1007/s11625-018-0627-5

Sætra, H.S., Danaher, J. (2022). To Each Technology Its Own Ethics: The Problem of Ethical Proliferation. *Philos. Technol. 35* (93). https://doi.org/10.1007/s13347-022-00591-7

Sætra, H.S., & Selinger, E. (2023). The siren song of technological remedies for social problems: Defining, demarcating, and evaluating techno-fixes and techno-solutionism. *SSRN Electronic Journal*. http://dx.doi.org/10.2139/ssrn.4576687

Saleebey, D. (1994). Culture, theory, and narrative: The intersection of meanings in practice. *Social Work, 39,* 351–359.

Sandler, R.L. (Ed.) (2014). *Ethics and Emerging Technologies*. New York: Palgrave Macmillan. https://doi.org/10.1057/9781137349088

Shilton, K. (2018). Values and Ethics in Human-Computer Interaction. *Foundations and Trends in Human-Computer Interaction, 12* (2), pp. 107–171.

Tepsa, T., Angelva, J., & Mielikäinen, M. (2023). Instructors' Expectations and Objectives for Integrating Sustainable Development and Ethical Issues Into the Curriculum. European Society for Engineering Education (SEFI). https://doi.org/10.21427/4ATQ-C251

Zhu, Q., Marin, L., Ramos, A.M. Sethy, S.S. (2025). The Purposes of Engineering Ethics Education. *In: Chance, S. et al. (Eds) The Routledge Interactional Handbook of Engineering Ethics Education.* London: Routledge. p. 27–43. https://doi.org/10.4324/9781003464259